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DUAL-CHAMBER DELIVERY DEVICE

Abstract

A drug delivery device configured to provide a dose of a first medicament followed by a dose of a second medicament is disclosed. The device includes a main exterior body and an interior body. The device further includes a dual-chamber reservoir with (i) a first chamber holding the first medicament and (ii) a second chamber holding the second medicament. The device also includes a needle and a plunger. After activation of the device, the plunger is configured to (i) move the dual-chamber reservoir a given distance in a distal direction with respect to the interior body so that a distal end of the needle extends out of the interior body and (ii) after moving the dual-chamber reservoir the given distance, move in the distal direction with respect to the dual-chamber reservoir so as to eject from the needle the first medicament followed by the second medicament.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/362,031 filed Jul. 31, 2023, which is a continuation of U.S. patent application Ser. No. 17/968,231 filed Oct. 18, 2022, which is a continuation of U.S. patent application Ser. No. 16/337,678 filed Mar. 28, 2019, which is a U.S. National Phase Application pursuant to 35 U.S.C. § 371 of International Application No. PCT/EP2017/074989 filed Oct. 2, 2017, which claims priority to U.S. Provisional Patent Application No. 62/404,868 filed Oct. 6, 2016. The entire disclosure contents of these applications are herewith incorporated by reference into the present application.

TECHNICAL AREA
[0002] This present disclosure relates to drug delivery devices such as automatic injection devices.
BACKGROUND
[0003] Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art by inclusion in this section.
[0004] In some situations, it is desirable for patients to be able to administer drugs and medicament by themselves, e.g., without the need for trained medical staff to administer the drugs. Further, in some situations, it is desirable to administer a first medicament followed by a second medicament in a single injection. There are a number of different existing delivery devices with varying degrees of automatic functions. For instance, existing automatic injection devices provide a means for automatically propelling a plunger forward to eject medicament from the automatic injection device in response to activation of the device. Further, existing automatic injection devices provide a means for ejecting a dose of a first medicament and a dose of a second medicament.
[0005] In existing devices, the means for penetrating an injection site and injecting a dose of a first medicament and a dose of a second medicament are often complex and expensive to manufacture. There is, therefore, a desire to reduce the cost of manufacturing automatic injection devices while maintaining the reliability of the injection device to penetrate an injection site and inject a dose of a first medicament followed by a dose of a second medicament.
SUMMARY
[0006] A dual-chamber drug delivery device is provided. In an example embodiment, the drug delivery device includes a main exterior body and an interior body axially moveable with respect to the main exterior body. Further, the drug delivery device includes a dual-chamber reservoir arranged in the interior body and axially moveable with respect to the interior body. The dual-chamber reservoir includes (i) a first chamber holding a first medicament and (ii) a second chamber holding a second medicament. Still further, the drug delivery device includes a plunger and a needle in fluid communication with the dual-chamber reservoir. Axial movement of the interior body in a proximal direction with respect to the main exterior body activates the drug delivery

device. Further, after activation of the drug delivery device, the plunger is configured to (i) move the dual-chamber reservoir a given distance in a distal direction with respect to the interior body so that a distal end of the needle extends out of the interior body and (ii) after moving the dual-chamber reservoir the given distance, move in the distal direction with respect to the dual-chamber reservoir so as to eject from the needle the first medicament followed by the second medicament.

[0007] In another example embodiment, the drug delivery device includes a main exterior body and an interior body axially moveable in a proximal direction with respect to the main exterior body to activate the drug delivery device. The drug delivery device further includes a dual-chamber reservoir arranged in the interior body and axially moveable with respect to the interior body. The dual-chamber reservoir includes a (i) first stopper slidably arranged in the dual-chamber reservoir and having a ruptureable membrane and (ii) a second stopper slidably arranged in the dual-chamber reservoir. A first chamber holding a first medicament is located between the first stopper and a distal end of the reservoir, and a second chamber holding a second medicament is located between the first stopper and the second stopper. Still further, the drug delivery device includes a needle in fluid communication with the first chamber and axially fixed to the first stopper. Axial movement of the interior body in a proximal direction with respect to the main exterior body activates the drug delivery device. Further, after activation of the drug delivery device, the plunger is configured to (i) move the dual-chamber reservoir a given distance in a distal direction with respect to the interior body so that a distal end of the needle extends out of the interior body and (ii) after moving the dual-chamber reservoir the given distance, move in the distal direction with respect to the dual-chamber reservoir so as to (a) eject from the needle substantially all of the first medicament, (b) subsequently rupture the ruptureable membrane to establish fluid communication between the needle and the second chamber, and (c) subsequently eject from the needle substantially all of the second medicament.

[0008] In yet another example embodiment, the drug delivery device includes a main exterior body and an interior body axially moveable in a proximal direction with respect to the main exterior body to activate the drug delivery device. The drug delivery device further includes a dual-chamber reservoir arranged in the interior body and axially moveable with respect to the interior body. The dual-chamber reservoir includes: (i) a first stopper slidably arranged in the dual-chamber reservoir; (ii) a second stopper slidably arranged in the dual-chamber reservoir, wherein a first chamber holding a first medicament is located between the first stopper and a distal end of the reservoir, wherein a second chamber holding a second medicament is located between the first stopper and the second stopper; and (iii) a conduit disposed in the first chamber, wherein the conduit comprises at least one conduit needle configured to establish fluid communication with the second chamber during an injection process. Still further, the drug delivery device includes a needle in fluid communication with the first chamber and axially fixed to the first stopper. Axial movement of the interior body in a proximal direction with respect to the main exterior body activates the drug delivery device. Further, after activation of the drug delivery device, the plunger is configured to (i) move the dual-chamber reservoir a given distance in a distal direction with respect to the interior body so that a distal end of the needle extends out of the interior body and (ii) after moving the dual-chamber reservoir the given distance, move in the distal direction with respect to the dual-chamber reservoir so as to (a) eject from the needle substantially all of the first medicament, (b) subsequently force the at least one conduit needle to fully penetrate the first stopper and establish fluid communication with the second chamber, and (c) subsequently eject from the needle substantially all of the second medicament.

[0009] In still yet another example embodiment, a method in a dual-chamber drug delivery device is provided. The dual-chamber drug delivery device includes (i) a body, (ii) a dual-chamber reservoir, (iii) a first stopper slidably arranged in the dual-chamber reservoir, (iv) a second stopper slidably arranged in the dual-chamber reservoir, wherein a first chamber holding a first medicament is between the first stopper and a distal end of the reservoir, and wherein a second chamber holding a second medicament is between the first stopper and the second stopper, (v) a needle in fluid communication with the dual-chamber reservoir, and (vi) a plunger. The method includes, in response to activation of the drug delivery device, the plunger moving the dual-chamber reservoir through the body of the device to force the needle out of the body and to penetrate an injection site. The method further includes, after moving the dual-chamber reservoir through the body of the device to expose the needle and penetrate an injection site, the plunger moving through the dual-chamber reservoir to force out of the drug delivery device substantially all the first medicament from the first chamber followed by substantially all of the second medicament from the second chamber.

[0010] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the figures and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Exemplary embodiments are described herein with reference to the drawings, in which:

[0012] FIG. **1** illustrates a perspective view of an example drug delivery device having an example dual-chamber reservoir, according to an example embodiment of the present disclosure.

[0013] FIG. **2** illustrates a partial cross-sectional view of the drug delivery device of FIG. **1** prior to removal of an example safety cap, according to an example embodiment of the present disclosure.

[0014] FIG. **3** illustrates a partial cross-sectional view of the drug delivery device of FIG. **1** after removal of the example safety cap of FIG. **2**, according to an example embodiment of the present disclosure.

[0015] FIG. **4** illustrates a cross-sectional view of the drug delivery device of FIG. **1** prior to needle penetration and medicament injection, according to an example embodiment of the present disclosure.

[0016] FIG. **5** illustrates a cross-sectional view of the drug delivery device of FIG. **1** after needle penetration and prior to medicament injection, according to an example embodiment of the present disclosure.

[0017] FIG. **6** illustrates a cross-sectional view of example components of the drug delivery device of FIG. **1**, according to an example embodiment of the present disclosure.

[0018] FIG. **7** illustrates a cross-sectional view of the drug delivery device of FIG. **1** after the plunger is released for medicament injection, according to an example embodiment of the present disclosure.

[0019] FIG. **8** illustrates a cross-sectional view of the drug delivery device of FIG. **1** after substantially all of a first medicament has been ejected from the drug delivery device, according to an example embodiment of the present disclosure.

[0020] FIG. **9** illustrates a cross-sectional view of the drug delivery device of FIG. **1** after substantially all of a second medicament has been ejected from the drug delivery device, according to an example embodiment of the present disclosure.

[0021] FIG. **10** illustrates a perspective view of an example drug delivery device having an example dual-chamber reservoir, according to an example embodiment of the present disclosure.

[0022] FIG. **11** illustrates a partial cross-sectional view of the drug delivery device of FIG. **10**, according to an example embodiment of the present disclosure.

[0023] FIG. **12** illustrates an example method, according to an example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0024] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.
[0025] The methods and systems in accordance with the present disclosure beneficially provide improved methods and systems for automatically ejecting a first and second medicament from an automatic injection device. The disclosed methods and systems provide a reliable, intuitive, and user-friendly drug delivery device that includes improved means for automatically propelling a plunger forward for needle penetration and injecting two medicaments one after the other. Further, the disclosed methods and systems provide cost effective means for needle penetration and injecting two medicaments one after the other and thus help to reduce the cost of manufacturing automatic injection devices.
[0026] In accordance with an example embodiment of the present disclosure, a drug delivery device includes a main exterior body and an interior body axially moveable with respect to the main exterior body. The drug delivery device further includes a dual-chamber reservoir arranged in the interior body and axially moveable with respect to the interior body. The dual-chamber reservoir includes (i) a first chamber holding a first medicament and (ii) a second chamber holding a second medicament. Still further, the drug delivery device includes a plunger, as well as a needle in fluid communication with the dual-chamber reservoir. The drug delivery device is configured such that axial movement of the interior body in a proximal direction with respect to the main exterior body activates the drug delivery device. Further, after activation of the drug delivery device, the plunger is configured to move the dual-chamber reservoir a given distance in a proximal direction with respect to the interior body so that a proximal end of the needle extends out of the interior body. Still further, the plunger is configured to, after moving the dual-chamber reservoir the given distance, move in the proximal direction with respect to the dual-chamber reservoir so as to eject from the needle the first medicament followed by the second medicament.
[0027] FIG. **1** generally illustrates an example drug delivery device that includes a dual-chamber sequential injection mechanism that allows it to inject two medicaments one after the other. In particular, FIG. **1** illustrates a drug delivery device **100** in an initial state prior to injection. Drug delivery device **100** includes a main exterior body **102** and interior body **104**. Interior body **104** is axially moveable with respect to main exterior body **102**. Exterior body **102** surrounds the majority of the interior body **104**; however, a proximal end **105** of the interior body **104** extends beyond a proximal end **107** of the main exterior body **102**. In the example of FIG. **1**, drug delivery device **100** also includes safety cap **106**.
[0028] With reference to FIG. **4**, drug delivery device **100** also includes a dual-chamber reservoir **108**, a needle **110** in fluid communication with the dual-chamber reservoir **108**, and a plunger **112**. The dual-chamber reservoir **108** is arranged in the interior body **104** and is axially moveable with respect to the interior body **104**. Further, the dual-chamber reservoir **108** includes (i) a first chamber **114** holding a first medicament **116** and (ii) a second chamber **118** holding a second medicament **120**. Drug delivery device **100** also includes a first stopper **115** slidably arranged in the dual-chamber reservoir **108** and a second stopper **119** slidably arranged in the dual-chamber reservoir **108**. First stopper **115** and second stopper **119** are in sliding fluid-tight engagement with the inner walls of dual-chamber reservoir **108**. The first chamber **114** is between the first stopper **115** and a proximal end **121** of the reservoir, and the second chamber **118** is between the first stopper **115** and the second stopper **119**.

[0029] As will be explained in greater detail below, axial movement of the interior body **104** in a distal direction **122** with respect to the main exterior body **102** activates the drug delivery device **100**. Further, after activation of the drug delivery device **100**, the plunger **112** is configured to (i) move the dual-chamber reservoir **108** a given distance **123** in a proximal direction **124** with respect to the interior body **104** so that a proximal end **126** of the needle **110** extends out of the interior body **104** and (ii) after moving the dual-chamber reservoir **108** the given distance **123**, move in the proximal direction **124** with respect to the dual-chamber reservoir **108** so as to eject from the needle **110** the first medicament **116** followed by the second medicament **120**. When plunger **112** moves in the proximal direction **124** with respect to the dual-chamber reservoir **108**, the plunger **112** will move axially through the reservoir **108** so as to push the stoppers **115**, **119** for medicament injection.
[0030] In the example embodiment of FIGS. **1-9**, dual-chamber reservoir **108** is a dual-chamber syringe. However, any suitable type of medicament reservoir may be used in the disclosed drug delivery device **100**, such as a syringe, an ampoule, a cartridge, an enclosure, etc. Further, the first medicament **116** and the second medicament **120** may be any suitable substances used for medical treatment. In an example embodiment, the first medicament is different than the second medicament. In another example embodiment, the first and second medicaments are both prescription drugs. Other medicaments are possible as well.
[0031] Operation of drug delivery device **100** is described in greater detail with reference to FIGS. **2-9**. In particular, an example safety-cap mechanism is described with reference to FIGS. **2-3**. Further, activation of the drug delivery device **100** is described with reference to FIG. **4** and needle penetration is described with reference to FIG. **5**. Still further, medicament injection is described with reference to FIGS. **7-9**.
[0032] FIG. **2** illustrates a partial cross-sectional view of the drug delivery device **100** prior to removal of safety cap **106**. As shown in FIG. **2**, drug delivery device **100** includes a biasing member **130**. Biasing member **130** is configured to bias the plunger **112** in proximal direction **124**. Prior to activation of drug delivery device **100**, the biasing member **130** is in a compressed state and continuously pushes against the plunger **112** as well as the inner surface of interior retainer **132**. At the same time, plunger arms **134**, **136** are held by a retaining washer **138**, which is disposed at a distal end of interior retainer **132**. As a result, the plunger **112** is prevented from moving axially away from the safety cap **106**. In particular, distal end portions **140**, **142** of plunger arms **134**, **136** are held by the retaining washer **138**. Further, before the safety cap **106** is removed, a safety pin **144** of safety cap **106** is located between the plunger arms **134**, **136** and thus prevents the plunger arms **134**, **136** from closing and escaping from the retaining washer **138**.
[0033] In this example embodiment shown in FIG. **2**, plunger **112** includes two plunger arms **134**, **136**. However, in other example embodiments, more or fewer plunger arms are possible. Further, in the example embodiment shown in FIG. **2**, biasing member **130** is a spring. However, other biasing members are possible as well. For instance, in another example embodiment, the biasing member is a pneumatic-based biasing member such as a pressurized gas source.
[0034] FIG. **3** illustrates a partial cross-sectional view of the drug delivery device **100** after removal of safety cap **106**. As seen in FIG. **3**, the safety cap **106** may be moved in distal direction **122** relative to the main exterior body **102**, so as to be removed from the drug delivery device **100**. After removal, the safety pin **144** of safety cap **106** is no longer located between the plunger arms **134**, **136**. Thus, the plunger arms **134**, **136** can later be closed together during the activation process so as to be narrow enough for escaping from the retaining washer **138**.
[0035] FIG. **4** illustrates a cross-sectional view of drug delivery device **100** during the activation process and prior to trigger needle penetration and medicament injection. As mentioned above, axial movement of the interior body **104** in a distal direction **122** with respect to the main exterior body **102** activates the drug delivery device **100**. This activation may occur when a user presses the drug delivery device **100** against an injection site, such as injection site **146**. The pressing force

pushes the interior body **104** against the interior retainer **132** which in turns brings the interior body **104**, retaining washer **138**, and the attached plunger **112** toward the exterior retainer **148**. Such movement of the interior body **104**, retaining washer **138**, and the attached plunger **112** toward the exterior retainer **148** can be seen by comparing FIG. **3** with FIG. **4**.

[0036] When the end portions **140**, **142** of plunger arms **134**, **136** reach the opening **150** of exterior retainer **148**, the opening **150** forces the end portions **140**, **142** as well as the whole plunger arms **134**, **136** to bend inward and have a diameter smaller than that of the opening on the retaining washer **138**. The biasing member **130** can then release stored energy, so as to push the plunger **112** away from the retaining washer **138** and exterior retainer **148**. In particular, biasing member **130** extends distally and starts pushing the plunger **112** in proximal direction **124**.

[0037] As mentioned above, after activation of the drug delivery device **100**, the plunger **112** is configured to move the dual-chamber reservoir **108** a given distance **123** in a proximal direction **124** with respect to the interior body **104** so that a proximal end **126** of the needle **110** extends out of the interior body **104**. This action causes the needle **110** to penetrate the injection site **146** to a desired depth. The plunger **112** is also configured to, after moving the dual-chamber reservoir **108** the given distance **123**, move in the proximal direction **124** with respect to the dual-chamber reservoir **108** so as to eject from the needle **110** the first medicament **116** followed by the second medicament **120**. This injection operation of (i) needle penetration to a desired depth followed by (ii) medicament delivery is described further with respect to FIGS. **5-9**.

[0038] FIG. **5** illustrates a cross-sectional view of drug delivery device **100** after needle penetration and prior to medicament injection. As seen in FIG. **5**, the plunger **112** has been released from the retaining washer **138** and is pushed by biasing member **130** in proximal direction **124**. Initially, when the plunger **112** is released, the biasing member **130** starts to push both the plunger **112** as well as the dual-chamber reservoir **108** toward the proximal end **105** of interior body **104**. The dual-chamber reservoir **108** travels the given distance **123** relative to the inner body **104** such that needle penetration into injection site **146** takes place.

[0039] Interior body **104** includes a stop feature that limits travel of the dual-chamber reservoir **108** in distal direction **108**. For instance, interior body **104** includes a reduced diameter portion **149** that acts as the stop feature and limits the proximal travel of the dual-chamber reservoir **108**. The reduced diameter portion **149** may include one or more protrusions, such as protrusions **151** and **153** (see FIG. **4**). In an example embodiment, the force required to move the dual-chamber reservoir **108** through the interior body **104** to the stop feature is less than the force required to move the second stopper **119** through the dual-chamber reservoir **108**. Thus, the dual-chamber reservoir **108** will move proximally through the interior body **104** to the stop feature prior second stopper **119** moving through the dual-chamber reservoir **108**. This will ensure that that needle **110** will be extended a desired amount prior to the ejection of any medicament through the needle **110**.

[0040] Returning to FIG. **5**, drug delivery device **100** includes a collapsible needle cover **154** covering the needle **110**. The collapsible needle cover **154** is configured to collapse and be penetrated by the needle **110** when the plunger **112** moves the dual-chamber reservoir **108** the given distance **123**. During this movement, the needle cover **154** is squashed as cover **154** is pushed against the interior body **104**, and then the needle **110** penetrates both the needle cover **154** and injection site **146**.

[0041] In an example embodiment, drug delivery device **100** includes a stopper grabber **156** that evenly distributes the force from the plunger **112** to the second stopper **119** during medicament injection. FIG. **6** illustrates the assembly of plunger **112**, stopper grabber **156**, and stoppers **115**, **119**, which are all moved by biasing member **130** for medicament injection. In an example embodiment, the stopper grabber **156** includes several ribs (not shown) to be in contact with the inner surface of the inner walls of the dual-chamber reservoir **108** in order to maintain axial location of the assembly during medicament injection, by evenly distributing the force from the biasing member **130** and plunger **112** to the second stopper **119**. In another example embodiment, the plunger **112** acts directly on second stopper **119** rather than through a stopper grabber **156**.

[0042] FIG. **7** illustrates a cross-sectional view of the drug delivery device **100** after the plunger **112** is released from the retainers **132**, **148** and driven by the biasing member **130** to apply force on the stopper grabber **156**. After the needle penetration takes place, the biasing member **130** continuously pushes the plunger **112**, stopper grabber **156**, and second stopper **119** through the dual-chamber reservoir **108** toward the needle **110** in order to force the medicament out of the needle and into the injection site. In particular, after the plunger **112** moves the dual-chamber reservoir **108** the given distance **123** for needle penetration, (i) force from the plunger **112** is transferred to the second stopper **119**, (ii) the second stopper **119** pushes the second medicament **120** in the second chamber **118**, (iii) the second medicament **120** consequently pushes the first stopper **115**, and (iv) the first stopper **115** consequently pushes the first medicament **116** in the first chamber **114** out of the needle **110**.

[0043] In an example embodiment, the needle **110** is axially fixed to the first stopper **115**. Therefore, as first stopper **115** moves axially through the dual-chamber reservoir **108** to eject to the first medicament **116**, the needle **110** further penetrates the injection site (see FIG. **7** compared to FIG. **8**). In another example embodiment, rather than being axially fixed to the first stopper, the needle **110** is axially fixed to a distal end of the dual-chamber reservoir **108**.

[0044] As seen in FIG. **7**, drug delivery device **100** includes a ruptureable membrane **160** separating the first chamber **114** from the second chamber **118**. The ruptureable membrane **160** is configured to rupture due to a pressure imbalance between the first chamber **114** and the second chamber **118** that occurs when substantially all the first medicament **116** in the first chamber **114** is ejected. During the injection of the first medicament **116**, the stopper grabber **156** transfers force to the second stopper **119** which in turn pushes the medicament **120** in the second chamber **118**, which in turns pushes the first stopper **115** and then the medicament **116** in the first chamber **114** out of the needle **110** and into the injection site **146**. At this stage of ejecting the first medicament **116**, there are medicaments **116**, **120** in both chambers which maintain a pressure balance between both chambers **114**, **118**. Thus, the membrane **160** can continue separating the two chambers from one another without rupture.

[0045] However, subsequent to ejecting all or substantially all of the first medicament **116**, the ruptureable membrane **160** ruptures to establish fluid communication between the needle **110** and the second chamber **118**. FIG. **8** illustrates drug delivery device **100** after substantially all of a first medicament **116** has been ejected from the drug delivery device **100** into the injection site **146** in full. At this stage, there is now a pressure imbalance between the first chamber **114** and the second chamber **118**. The second medicament **120** in the second chamber **118**, under forces from the biasing member **130**, continues pushing and eventually ruptures the membrane **160** shortly after the first chamber **114** is emptied or substantially emptied.

[0046] After this rupture, the second medicament **120** is then ejected from the drug delivery device **100**. In particular, after the membrane **160** ruptures, the second medicament **120** can flow into the first chamber **114**. The second medicament **120** previously in the second chamber **118** is then pushed by the biasing member **130** and exits the dual-chamber reservoir **108** through the needle **110** in the first chamber **114** and into the injection site **146**. FIG. **9** illustrates a drug delivery device **100** after substantially all of the second medicament **120** has been ejected.

[0047] In an example embodiment, the first stopper **115** comprises the ruptureable membrane **160**. For instance, the first stopper **115** may be manufactured (e.g., via injection molding) to include the ruptureable membrane **160**. In another example embodiment, the membrane **160** is attached to the first stopper **115** in any suitable fashion. For example, the membrane **160** may be attached to the first stopper **115** with an adhesive.

[0048] In an example embodiment, during dose delivery, the user can hear and/or feel an audible and/or tactile feedback (e.g., clicking) throughout the dose delivery. For instance, the drug delivery device **100** may include a clicker that produces a clicking sound when the plunger **112** is being propelled forward in the proximal direction **124**. The end of injection may be indicated by the audible/tactile clicking having stopped. Additionally or alternatively, the drug delivery device **100** may include a window that allows a user to see when injection is complete. In such an example where the drug delivery device **100** includes the window, the end of delivery is indicated by the second stopper **119** and plunger **112** having stopped moving. Other indications of dose delivery being complete are possible as well.

[0049] FIGS. **10** and **11** depict another example embodiment of the disclosed dual-chamber drug delivery device. In particular, FIGS. **10** and **11** depict drug delivery device **200**. The drug delivery device **200** operates in a similar fashion as drug delivery devices **100**; however, rather than including ruptureable membrane **160**, drug delivery device **200** includes a conduit disposed in the first chamber **114**. The conduit is configured to establish fluid communication with the second chamber **118** during the injection process. Further, the conduit comprises at least one conduit needle configured to penetrate the first stopper **115** and establish the fluid communication with the second chamber **118**. Other elements of drug delivery device **200** may be the same or substantially similar to the other elements drug delivery device **100**, and thus drug delivery device **200** is not described in as great of detail. It should be explicitly noted, however, that any possibilities and permutations described above with respect to drug delivery device **100** may equally apply to drug delivery device **200**, and vice versa. Further, throughout the description of FIGS. **10-11**, elements in drug delivery device **200** that are the same as or substantially similar to elements in drug delivery device **100** are described with like reference numerals.

[0050] Turning to FIG. **11**, the needle **110** has a needle opening **202** that allows medicament to enter the needle **110** and then to be injected into the injection site **146**. On the other hand, the conduit **204** in the first chamber **114** includes two conduit needles **206**, **208**, with internal passages, that will penetrate the first stopper **115** when it is pushed by the medicament **120** in the second chamber **118** toward the first chamber **114**. Although two conduit needles **206**, **208** are shown, in other example embodiments, more or fewer conduit needles are possible.

[0051] During medicament injection, when the first stopper **115** pushes all or substantially all of the first medicament **116** in the first chamber **114** out through the needle **110**, the conduit needles **206**, **208** of conduit **204** will penetrate the first stopper **115** and enter the second chamber **118** to form a passage between the two chambers **114**, **118**. In this way, the medicament **120** in the second chamber **118** can then be pushed by the second stopper **119** (which is pushed by the stopper grabber **156**, plunger **112**, and biasing member **130**). The second medicament **120** then enters the first chamber **114**, enters the needle **110** through the needle opening **202**, and thereafter exits the needle **110**.

[0052] In an example embodiment, the conduit needles **206**, **208** are sized such that the needles **206**, **208** fully penetrate the first stopper **115** and establish the fluid communication with the second chamber **118** subsequent to all or substantially all of a first medicament **116** in the first chamber **114** being ejected. After the needles **206**, **208** fully penetrate the first stopper **115**, needle opening **202** will be located in a proximal portion **210** of the first chamber **114**. The second medicament **120** will flow through conduit needles **206**, **208** to the distal portion **210** of the first chamber **114** and then through the needle opening **202**, so as to be ejected from needle **110**.

[0053] FIG. **12** illustrates an example method **300** that can be carried out in a dual-chamber drug delivery device in accordance with the present disclosure, such as drug delivery device **100** or **200**. The method includes, at block **302**, in response to activation of the drug delivery device, the plunger moving the dual-chamber reservoir through the body of the device to force the needle out of the body and to penetrate an injection site. The method further includes, at block **304**, after moving the dual-chamber reservoir through the body of the device to expose the needle and penetrate an injection site, the plunger moving through the dual-chamber reservoir to force out of the drug delivery device substantially all the first medicament from the first chamber followed by substantially all of the second medicament from the second chamber.

[0054] In an example embodiment of method **300**, the plunger moving through the dual-chamber reservoir to force out of the drug delivery device substantially all the first medicament followed by substantially all of the second medicament involves: (i) force from the plunger transferring to the second stopper; (ii) the second stopper pushing the second medicament in the second chamber, (iii) the second medicament pushing the first stopper; (iv) the first stopper pushing the first medicament in the first chamber out of the needle; and (v) after the first stopper pushing the first medicament in the first chamber out of the needle, the second stopper pushing the second medicament in the second chamber out of the needle.
[0055] In an example embodiment of method **300**, the body of the drug delivery device is an interior body, such as interior body **104**. In another example of method **300**, however, the body of the drug delivery device is a main exterior body of the drug delivery device.
[0056] In the examples shown in the Figures, drug delivery devices **100** and **200** are activated by being pressed against injection site **146** after removal of the safety cap **106**. However, in other example embodiments, the disclosed drug delivery device may be activated in other ways. In general, the drug delivery device may be activated in any suitable fashion. For instance, in an example embodiment, the drug delivery device includes a push button and the drug delivery device is activated by a combination of the drug delivery device being pressed against an injection site and the push button being depressed by a user. In another example embodiment, the drug delivery device is activated by a push button being depressed by a user. Other types of activation are possible as well.
[0057] Further, in the examples shown in the Figures, the exterior retainer **148** is axially fixed to main exterior body **102**. In an example embodiment, these elements are manufactured separately and then axially fixed during an assembly of the drug delivery device. However, in another example, these elements may be of unitary construction. Similarly, in the examples shown in the Figures, interior retainer **132** and interior body **104** are manufactured separately and then assembled in the drug delivery device **100**. However, in another example, these elements may be of unitary construction.
[0058] In the Figures, various engagement features for are shown for providing an engagement between one or more components of the drug delivery device. The engagement features may be any suitable connecting mechanism such as a snap lock, a snap fit, form fit, a bayonet, lure lock, threads or combination of these designs. Other designs are possible as well.
[0059] Throughout the description, by the term “substantially” it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide. For instance, in an example embodiment, substantially all of the medicament being ejected from a chamber means at least 95% of the medicament originally held in the chamber is ejected from the chamber. In another example embodiment, substantially all of the medicament being ejected from a chamber means at least 90% of the medicament originally held in the chamber is ejected from the chamber.
[0060] Beneficially, the disclosed dual-chamber drug delivery device and disclosed method provide a cost effective means for propelling a plunger forward to penetrate an injection site and eject a dose of a first medicament followed by a dose of a second medicament. Therefore, the disclosed dual-chamber drug delivery device may help to reduce the cost of manufacturing automatic injection devices.
[0061] It should be understood that the illustrated components are intended as an example only. In other example embodiments, fewer components, additional components, and/or alternative components are possible as well. Further, it should be understood that the above described and shown embodiments of the present disclosure are to be regarded as non-limiting examples and that they can be modified within the scope of the claims.

[0062] While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope being indicated by the following claims, along with the full scope of equivalents to which such claims are entitled. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

Claims

1. (canceled)
2. A drug delivery device comprising: a main exterior body; an interior body axially moveable with respect to the main exterior body; a dual-chamber reservoir arranged in the interior body and axially moveable with respect to the interior body, the dual-chamber reservoir comprising (i) a first chamber holding a first medicament, and (ii) a second chamber holding a second medicament; a first stopper slidably arranged in the dual-chamber reservoir; a second stopper slidably arranged in the dual-chamber reservoir, wherein the first chamber holding the first medicament is located between the first stopper and a proximal end of the dual-chamber reservoir, and wherein the second chamber holding the second medicament is located between the first stopper and the second stopper; a needle in fluid communication with the dual-chamber reservoir; and a plunger coupled to the second stopper, wherein activation of the drug delivery device occurs when a user presses a proximal end of the interior body against an injection site thereby causing the plunger to move the first stopper in a proximal direction with respect to the interior body from a first position to a second position.
3. The drug delivery device of claim 2, wherein the dual-chamber reservoir is a dual-chamber syringe.
4. The drug delivery device of claim 2, further comprising a biasing member configured to bias the plunger in the proximal direction after the activation of the drug delivery device.
5. The drug delivery device of claim 4, wherein the biasing member is a spring.
6. The drug delivery device of claim 2, wherein the second medicament can only flow into the first chamber and pass through the needle when the first stopper is in the second position.
7. The drug delivery device of claim 2, wherein a distal end of the plunger includes a plunger holding mechanism configured to hold the plunger in place prior to the activation of the drug delivery device.
8. The drug delivery device of claim 7, wherein the plunger holding mechanism comprises a pair of bendable arms.
9. The drug delivery device of claim 2, wherein a distal end of the drug delivery device includes a safety cap configured to hold the plunger in place prior to the activation of the drug delivery device.
10. The drug delivery device of claim 2, wherein an end of delivery is indicated when the second stopper and the plunger have stopped moving.
11. The drug delivery device of claim 2, wherein the interior body includes a stop feature that limits travel of the dual-chamber reservoir in a distal direction, and wherein a force required to move the dual-chamber reservoir through the interior body to the stop feature is less than a force required to move the second stopper through the dual-chamber reservoir such that the dual-chamber reservoir will move in the proximal direction through the interior body to the stop feature prior to the second stopper moving through the dual-chamber reservoir.
12. The drug delivery device of claim 11, wherein the stop feature comprises a reduced diameter portion of the interior body.
13. The drug delivery device of claim 12, wherein the reduced diameter portion includes one or more protrusions.

14. A method comprising: in response to activation of the drug delivery device of claim 2, the plunger moving the dual-chamber reservoir through the interior body of the drug delivery device; and the plunger moving through the dual-chamber reservoir to (i) force out of the drug delivery device substantially all the first medicament from the first chamber, and (iii) subsequently force out of the drug delivery device substantially all of the second medicament from the second chamber.
15. The method of claim 14, the method further comprising: after being pressed against the injection site, the interior body moves in a distal direction with respect to the main exterior body to activate the drug delivery device.
16. The method of claim 14, wherein the plunger moving through the dual-chamber reservoir to force out of the drug delivery device substantially all the first medicament followed by substantially all of the second medicament comprises: force from the plunger transferring to the second stopper; the second stopper pushing the second medicament in the second chamber, the second medicament pushing the first stopper; the first stopper pushing the first medicament in the first chamber out of the needle; and after the first stopper pushing the first medicament in the first chamber out of the needle, the second stopper pushing the second medicament in the second chamber out of the needle.
17. The method of claim 14, wherein the second medicament can only flow into the first chamber and pass through the needle when the first stopper is in the second position.